

Toward Visually Robust Quadruped Parkour via Monocular Depth Estimation

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MTHE 499 — Supervisor: Prof. Joshua Marshall

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Personal Motivation

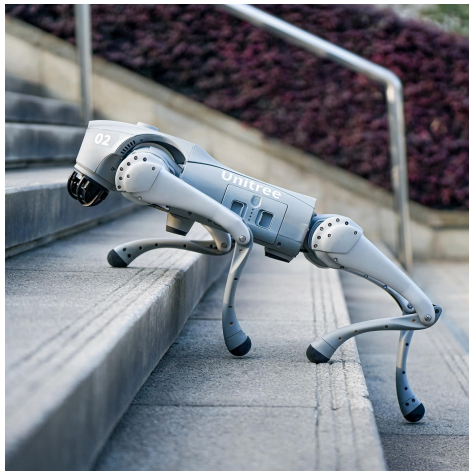
- Programmer on high school robotics team
- Tried to understand Kalman filters, LQR, optimal control, etc, but couldn't follow the math
- That's why I chose Apple Math!
- At the end of my degree, wanted one last project to actually use what I learned
- So I bought a robot dog from China. . .



Problem Definition

What is Quadruped Parkour?

- A legged robot autonomously traversing obstacles using **vision** and **agile control**.
- Obstacles include stairs, hurdles, and gaps.
- Policies are trained in simulation and transferred to hardware.
- Control 12 joints at 50 Hz from partial observations.
- Difficult because simulation-to-reality gap is large for vision-based control



Prior Work: Vision-Based Parkour

- **Scandots** [Agarwal et al. 2023]
Heightmap teacher → depth-camera student
- **Extreme Parkour** [Cheng et al. 2023]
CNN-GRU student; real-robot deployment
- **Robot Parkour Learning** [Zhuang et al. 2023]
Multi-skill parkour on hardware
- **ANYmal Parkour** [Hoeller et al. 2024]
Hierarchical skill selection

All assume a depth camera at deployment.

Agarwal et al. 2023 | Cheng et al. 2023 | Zhuang et al. 2023 | Hoeller et al. 2024

Our Question

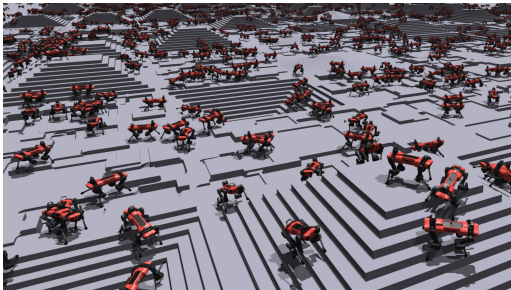
Can **monocular depth estimation** replace a depth camera for quadruped parkour?



- Unitree Go2: RGB camera only
- Goal: train in estimated depth space, deploy from RGB

Background

Simulation Infrastructure



Legged Gym [Rudin et al. 2022]

Massively parallel sim-to-real RL framework

Rudin et al. 2022 | Genesis, 2024

Genesis simulator

add: pictures/genesis-sim.jpg

Genesis GPU-accelerated physics simulator

Bridging the Sim-to-Real Gap

1. **Domain Randomization** [Tobin et al. 2017]
Randomize textures, lighting, and physics during training
2. **Generative Augmentation** [Yu et al. 2024 — LucidSim]
Diffusion models for photorealistic images
3. **Canonical Representation**
Project both sim and real into a **shared depth space** via a pre-trained foundation model

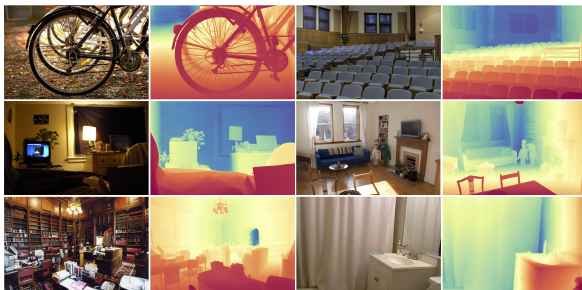


LucidSim: diffusion-rendered training images (Yu et al. 2024)

Tobin et al. 2017 | Yu et al. 2024 (LucidSim) | Yang et al. 2024 (DA2)

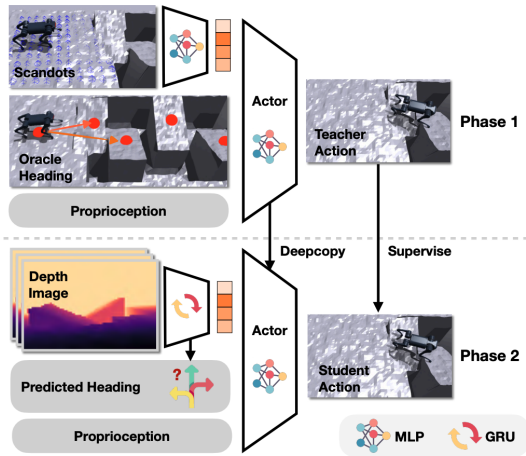
Depth Anything V2 (DA2)

- Vision Transformer (ViT)
- Trained on 142M real images
- RGB $\rightarrow$ metric depth map
- Multiple model scales:
small, base, large



Yang et al., arXiv 2024

Two-Stage Distillation Pipeline



Mathematical Framework

Problem Formulation

Partially Observable Markov Decision Process $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, R, \gamma)$

Observations

- Proprioception $o_{\text{prop}} \in \mathbb{R}^{53}$
angular vel., gravity, joints, contacts
- Exteroception o_{ext}
scandots $\mathbb{R}^{132}$ or depth image

Actions

- Joint offsets $a_t \in \mathbb{R}^{12}$
- PD controller: $\tau = K_p(a_t - q) - K_d\dot{q}$

Objective

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right]$$

PPO

Clipped surrogate objective

$$L^{\text{CLIP}} = \mathbb{E}_t \left[\min \left(r_t \hat{A}_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon) \hat{A}_t \right) \right]$$

$$r_t = \pi_{\theta}(a_t|o_t) / \pi_{\theta_{\text{old}}}(a_t|o_t)$$

Clip prevents large, destructive updates

Generalized Advantage Estimation (GAE)

$$\hat{A}_t^{\text{GAE}(\gamma, \lambda)} = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}, \quad \delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$$

λ interpolates: $\lambda=0$ (low variance, high bias) $\leftrightarrow$ $\lambda=1$ (high variance, low bias)

Schulman et al., arXiv 2017; Schulman et al., ICLR 2016

Dagger Distillation

Student acts, teacher labels, iterate:

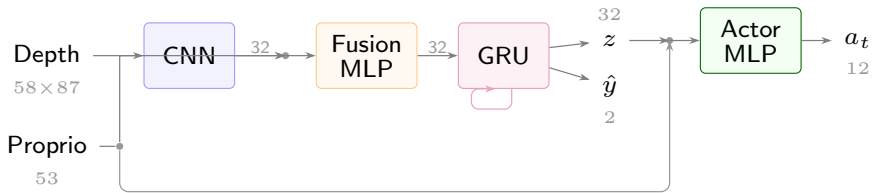
$$\mathcal{L} = \underbrace{\|a_{\text{teacher}} - a_{\text{student}}\|_2}_{\text{action loss}} + \underbrace{\|y_{\text{oracle}} - \hat{y}\|_2}_{\text{heading loss}}$$

The student runs in the environment and collects its own observations, but the teacher's actions are executed.

Ross et al., AISTATS 2011

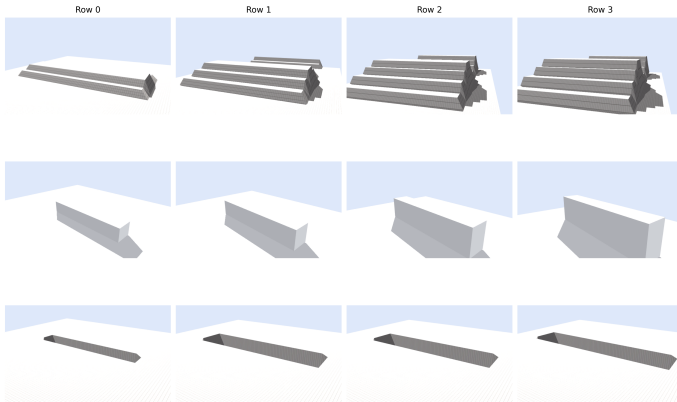
System Architecture

Student Visual Encoder



Depth CNN → Fusion MLP → GRU → terrain latent $z \in \mathbb{R}^{32}$ + heading $\hat{y} \in \mathbb{R}^2$

Terrain and Curriculum



Three obstacle families $\times$ four difficulty levels.

Simulation Demo

Video demo — show simulation runs here

Experiments & Results

Baselines: Upper Bounds

	Row 0	Row 1	Row 2	Row 3
Scandots teacher	98.5	100.0	99.0	98.5
GT-depth student	94.5	94.5	89.1	87.5
Zero-depth ablation	8.0	2.0	0.0	0.0

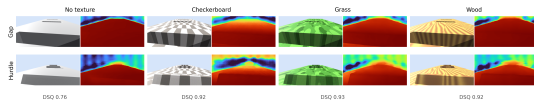
The student architecture works.

Depth quality is the bottleneck, not the network.

Estimated Depth: Model Scale and Texture

	R0	R1	R2	R3
DA2-small	56	34	—	—
DA2-base	68	83	27	1
DA2-base+tex	65	64	68	45
GT-depth	95	95	89	88

Larger model + texture
→ closes gap to GT-depth



Depth Signal Quality (DSQ)

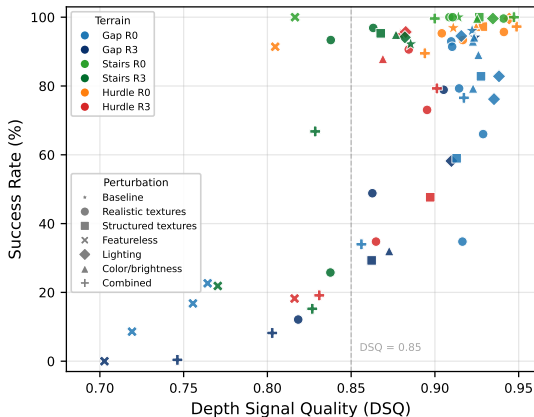
A diagnostic metric: how well does estimated depth preserve terrain geometry?

$$\text{DSQ} = \frac{1}{T} \sum_{t=1}^T \frac{1}{N} \sum_{i=1}^N \rho(d_i^{\text{GT}}, d_i^{\text{DA2}})$$

Pearson correlation is **affine invariant**: $\rho(\alpha x + \beta, y) = \rho(x, y)$

→ naturally handles monocular scale-shift ambiguity

Visual Sensitivity



- Stairs/hurdles: robust (strong geometric edges)
- Gaps: fragile (boundary depends on texture)
- Featureless → catastrophic
- Lighting/color → moderate

DSQ predicts failure well,
but high DSQ $\neq$ guaranteed success

Takeaways

What We Learned

1. Monocular depth is a **plausible bridge** from RGB to locomotion
2. Bottleneck = **depth quality**, not student architecture
3. Robust to photometric variation; **fragile on featureless surfaces**

Future work

- Real-world transfer on Unitree Go2
- Temporally consistent depth models
- Joint adaptation of depth estimator + student policy

Thank you!

Questions?